Manvendra Sang

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OBJECTIVE

Full-Stack Developer and AI/ML enthusiast with hands-on experience in MERN, Python, C/C++, and machine learning, including AI/ML alignment workflows, seeking a role where I can build web applications and AI-powered solutions.

SKILLS

Technical Skills: Python, JavaScript, TypeScript, C/C++, ReactJS, HTML5, Tailwind CSS, Node.js, CRM, HRM, CRUD apps

Databases & Cloud: MySQL, MongoDB, REST APIs, Google Cloud Platform, AWS

AI / ML: Deep Learning (MLP, CNN, RNN, Autoencoders), Reservoir Computing (ESN), LLM Fine-tuning, RAG, RLHF, MCP's, Query Modelling, Retrieval Quality and Modelling

Tools & DevOps: Git, Railway, Godot, Blender, Nvidia NIM, Hugging Face, Linux (Arch Linux, KDE Plasma, Hyrpland)

Certifications: MongoDB Python Developer, Data Structures (UC San Diego), NPTEL Java, NPTEL Cloud Computing, NPTEL Introduction to Internet of Things, Publication ICDAM 2024: Voice Recognition Systems

PROJECTS

Agentic AI, IT Ticket Manager: Building an AI agent that auto-resolves low-urgency support tickets and escalates only what requires human intervention, reducing workload significantly. Built in Python with modular OOP and LLM-based triage logic.

MERN Team Task Manager: Full-stack live application for team task assignment, urgency escalation, and warnings, hosted on Railway with full Git version control. Built with MongoDB, Express, ReactJS, Node.js.

EEG Phoneme Classifier (Reservoir Computing): AI model to classify brain-wave responses to phonemes using Echo State Networks, targeting communication aid for paralyzed/coma patients. Achieved 78.7% classification accuracy.

EXPERIENCE

Research Intern, Deep Learning & Reservoir Computing **Jan 2026 to Mar 2026**
Kyushu Institute of Technology (Tanaka Lab), Fukuoka, Japan

- Studied core deep learning architectures from fundamentals to advanced: MLP, CNN, RNN, and Autoencoders.
- Explored Reservoir Computing (Echo State Networks) theory and implementation under direct mentorship.
- Applied query modelling and retrieval quality analysis to improve signal-to-label mapping in the classification pipeline.
- Developed an Edge-Based EEG Speech Emotion Decoder leveraging reservoir computing, achieving a peak classification accuracy of 78.7%.

Research Intern, AI/ML & Deep Learning (Hybrid) **Jun 2025 to Sep 2025**
IIT Delhi, Hauz Khas, New Delhi

- Trained and fine-tuned LLMs via external API calls from Hugging Face and Grok.
- Fine-tuned models with target and translated language datasets, evaluating accuracy across iterations.
- Built RAG pipelines to ground model outputs in verified reference data, improving retrieval quality for low-resources.
- Applied RLHF to align fine-tuned model outputs with human-rated translation quality.
- Benchmarked model inference using Nvidia NIM microservices to evaluate deployment latency and throughput.

Intern, Full-Stack Development (Remote) **Sep 2025 to Oct 2025**
Spoken Tutorial, EduPyramids, SINE, IIT Bombay, Remote

- Developed and maintained features for the Spoken Tutorial website using full-stack web technologies.
- Migrated select components to TypeScript to improve type safety and reduce runtime errors.
- Managed codebase version control using Git, maintaining clean commit history and collaborative workflows.
- Worked with HTML, HTML5, and related frontend technologies as part of the EduPyramids/SINE team.

EDUCATION

Bachelor of Technology, 2022 to 2026
Dronacharya College of Engineering, New Delhi

Secondary School (CBSE Board), 2008 to 2022
CBSE BOARD, INDIA